

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-A51CC96085

Country: Tanzania

Submission Date: 2026-02-22

Product: Artesunex (acyclovir) | ATC: J05AB01 | Strength: 200 mg

Applicant: United Therapeutics | Manufacturer: Bobo-Dioulasso Pharma SA

Policy Label: reject_and_return | Risk Score: 0.92

AMR Stewardship: AwaRe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable
| Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

Application dossier for Artesunex (acyclovir) submitted on 2026-02-22 to the Tanzania authority. Applicant: United Therapeutics. Proposed dosage form: suspension; strength: 200 mg; ATC: J05AB01. Administrative declarations are present but naming conflict remediation documents are incomplete. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Administrative cover letters, declarations, and fee records were cross-checked against the submission index. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. The proposed prescribing information and patient information leaflet have been translated into local languages. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: gmp_certificate_expired

Primary manufacturing site: Bobo-Dioulasso Pharma SA in Tanzania. Latest GMP inspection date: 2025-11-24. GMP status: compliant. GMP certificate number: GMP-131244; certificate expiry: 2025-02-05. Certificate has expired and requires renewal before authorization. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Site quality-system commitments were compared against the proposed commercial manufacturing activities. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable

Site Inspection History Summary:

Inspection Date	Authority	Scope	Outcome
2025-11-24	National Auth	FPP/General	compliant
2022-05-12	SRA Joint	API/Sterile	compliant

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: inn_infringement

Proposed product name is Artesunex, with INN listed as acyclovir. Name similarity review indicates potential confusion with protected and existing nomenclature classes. The submission did not include a sufficient differentiation and risk mitigation plan for look-alike and sound-alike naming concerns. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. A

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for acyclovir. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. All referenced documents are available in the dossier annex and traceable by section identifier. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The applicant confirms alignment with applicable technical guidance and regional submission standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. All referenced documents are available in the dossier annex and traceable by section identifier. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable technical guidance and regional submission standards. Statistical significance

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: clinical_failed

Clinical overview for community acquired pneumonia indicates pivotal efficacy outcomes did not meet the primary endpoint. Current evidence does not support a favorable benefit-risk conclusion for authorization. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The target population demographics are broadly representative of the intended commercial patient cohort. No new safety signals were identified during the extended open-label follow-up period. All referenced documents are available in the dossier annex and traceable by section identifier. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Clinical interpretation focuses on whether efficacy, safety, and benefit-risk conclusions are supported by the submitted evidence. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The applicant confirms alignment with applicable technical guidance and regional submission standards. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Pharmacokinetic parameters were deri

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=present; length=length_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Batch analysis, validation, and impurity management records support the proposed API quality framework. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The synthetic route involves three

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Adverse events were coded using the Medical Dictio

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. A randomized, double-blind, parallel-group study design was

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound.

Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Data were analyzed using a mixed-effects model f

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (community acquired pneumonia). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in acc

Pivotal Clinical Trial Inventory:

Protocol ID	Phase	Enrollment	Status
PROT-001	III	360	completed
PROT-002	III	449	completed

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: clinical_failed

Pivotal trial analysis indicates primary endpoint was not met in the intent-to-treat population. Estimated treatment effect did not reach pre-specified significance and sensitivity analyses were inconsistent. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced across all study arms. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic

parameters were derived using non-compartmental analysis from the concentration-time profiles. The applicant confirms alignment with applicable technical guidance and regional submission standards. Method validation records and quality controls were reviewed for internal consistency. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) versi

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of C_{max} and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. All referenced documents are available in the dossier annex and traceable by section identifier. The risk management plan includes routine pharmacovigilance and proposed educational materials fo